DO NOW — RECALL

# Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

**1. Distinguish RAM from virtual storage.**

**2. In a BST, where is a value less than the current node placed?**

**3. Why can an array provide direct indexed access?**

**4. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

RAM is local working memory; virtual storage presents remote/pooled file storage.

### 2. Model answer

In its left subtree (under the chosen no-duplicate convention).

### 3. Model answer

Its element locations can be calculated from base address and index.

### 4. Model answer

Sum 0 and carry-out 1.

### By the end, you can…

• The OS creates and schedules processes; programs share CPU time.

• Memory, files and device access are mediated by the OS.

• Interfaces, user accounts and access controls make a computer usable.

### Vocabulary — know these terms

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

NEW LEARNING

# Read, annotate and explain

### Manage processes

The OS creates and schedules processes; programs share CPU time.

### Manage resources

Memory, files and device access are mediated by the OS.

### Provide services

Interfaces, user accounts and access controls make a computer usable.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A word processor saves a document while music is playing. Explain three OS services involved.**

### Model answer

The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected.

Check: link each claim to the scenario and use precise OCR terminology.

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Describe three functions of an operating system, linking each to a user action. [6]

### Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**The operating system only provides a graphical user interface.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.